

Mathematics for Data Science

Lecture 4 - Eigendecomposition

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Previously covered topics

- (Lecture 1) Vector spaces, subspaces, linear transformations. Rank, image, kernel.
- (Lecture 2) Matrices, link with linear transformations, linear systems.
- (Lecture 3) Determinant

In this lecture: Eigendecomposition.

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 - Eigendecomposition of an endomorphism
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Eigenvalues and eigenvectors

Definition

Let A be an $n \times n$ matrix. A nonzero vector $v \in \mathbb{R}^n$ is called an *eigenvector* of A if there exists $\lambda \in \mathbb{R}$ such that:

$$Av = \lambda v.$$

The scalar λ is called an *eigenvalue* of A associated with v .

Example

Find (i) eigenvalues of I_n and (ii) eigenvalues of $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

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Find (i) eigenvalues of I_n and (ii) eigenvalues of $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

- Any vector $x \in \mathbb{R}^n$ satisfies $I_n x = 1 \cdot x$, i.e. so every nonzero vector is an eigenvector of I_n associated with the eigenvalue 1.
- For $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, each canonical vector e_i is an eigenvector associated with the eigenvalue λ_i .

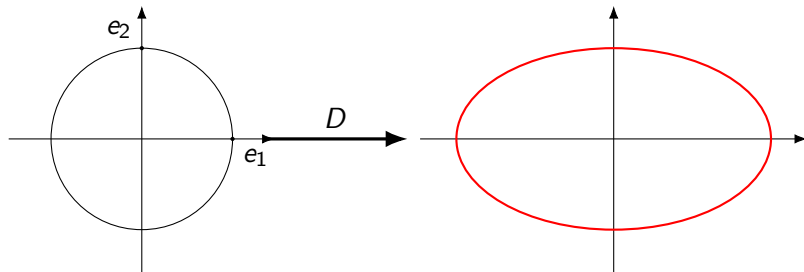
Geometrical interpretation

Example

In 2D : represent v an eigenvector of matrix A associated with eigenvalue λ . Discuss according to the value of λ .

In $\mathbb{R}^2$, how is the unit circle transformed via D ?

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{Unit circle } \left\{ x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mid x_1^2 + x_2^2 = 1 \right\}.$$



Geometrical interpretation

The circle is transformed into the red ellipse.

Let

$$y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = Dx \quad \text{with } x \in \text{unit circle (i.e. } x_1^2 + x_2^2 = 1), \quad y = \begin{pmatrix} 2x_1 \\ x_2 \end{pmatrix}.$$

Hence $x_1 = \frac{y_1}{2}$, $x_2 = y_2$, and substituting into $x_1^2 + x_2^2 = 1$ gives

$$\boxed{\frac{y_1^2}{4} + y_2^2 = 1} \quad (\text{Equation of an ellipse}).$$

Eigenspace and eigenspectrum

Definition (Eigenspace)

The **eigenspace** of A associated with an eigenvalue λ is the set of all eigenvectors associated with λ , together with the zero vector:

$$E_\lambda = \{v \in \mathbb{R}^n \mid Av = \lambda v\} = \ker(A - \lambda I_n)$$

Exercise : prove that it is a subspace.

NB : If a set of eigenvectors of A forms a basis of the domain of A , then this basis is called an **eigenbasis**.

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Definition (Eigenspectrum)

The **eigenspectrum** (or **spectrum**) of a matrix is the list of its eigenvalues, repeated according to their multiplicity.

NB: We will see that an important quantity associated with the spectrum is the maximum absolute value of any eigenvalue. This is known as the **spectral radius** of the matrix.

Diagonalization

Proposition

If A has n linearly independent eigenvectors $\{v_1, \dots, v_n\}$ with associated eigenvalues $\{\lambda_1, \dots, \lambda_n\}$, then these eigenvectors form a basis of $\mathbb{R}^n$ and A is **diagonalizable**:

$$A = VDV^{-1}$$

where $V = (v_1 \dots v_n)$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

NB: with the above notations, we can also write

$$AV = VD$$

Diagonalization

In other words, if $V \in \mathcal{M}_{n \times n}(\mathbb{R})$ is composed of n linearly independent vectors:

$$\text{span}(v_1, \dots, v_n) = \mathbb{R}^n \Rightarrow \text{rank}(V) = n \Rightarrow V \text{ is an invertible matrix.}$$

From $AV = VD$, if V is invertible, we get

$$AV = VD \iff AVV^{-1} = VDV^{-1} \iff A = V \underset{\text{Diagonal}}{D} V^{-1}.$$

We say that the matrix A is **diagonalizable**.

Computing AV in detail

$$AV = A(v_1 \cdots v_n) = (Av_1 \cdots Av_n) = (\lambda_1 v_1 \cdots \lambda_n v_n) = VD.$$

$$V = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}, \quad D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix},$$

$$VD = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = \begin{bmatrix} | & & | \\ \lambda_1 v_1 & \cdots & \lambda_n v_n \\ | & & | \end{bmatrix}.$$

$$(VD)_{ij} = \sum_{k=1}^n V_{ik} D_{kj} = V_{ij} \lambda_j, \quad (AV)_{ij} = (Av_j)_i = (\lambda_j v_j)_i = V_{ij} \lambda_j,$$

hence $AV = VD$.

Notes around $AV = VD$

Warning

$$VD \neq DV$$

- VD multiplies the columns of V by the diagonal entries d_i .
- DV multiplies the rows of V by the diagonal entries d_i .

Exercise: Powers of a matrix with a basis of eigenvectors

Example

Let $\{(\lambda_i, v_i)\}_{i=1}^n$ be eigenvalue–eigenvector pairs of a matrix $A \in \mathbb{R}^{n \times n}$ such that $\{v_1, \dots, v_n\}$ are linearly independent. Identify the eigenvalues and eigenvectors of $A^2, \dots, A^k$ for $k \in \mathbb{N}^*$. Are these matrices diagonalizable?



Exercise: Powers of a matrix with a basis of eigenvectors

Since $Av_i = \lambda_i v_i$ and the v_i 's form a basis, we have, by induction on $k > 0$,

$$A^k v_i = \lambda_i^k v_i \quad \text{for } i = 1, \dots, n, \quad k \in \mathbb{N}^*.$$

Hence for every $k \in \mathbb{N}$ the vectors $v_1, \dots, v_n$ are eigenvectors of A^k and the corresponding eigenvalues are $\lambda_1^k, \dots, \lambda_n^k$.

Because the same n eigenvectors remain linearly independent, A^k is diagonalizable: if $V = (v_1 \cdots v_n)$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, then

$$A = VDV^{-1} \implies A^k = VD^kV^{-1} = V \text{diag}(\lambda_1^k, \dots, \lambda_n^k) V^{-1}.$$

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$$A = VDV^{-1} \implies A^k = VD^kV^{-1} = V \text{diag}(\lambda_1^k, \dots, \lambda_n^k) V^{-1}.$$

Remarks.

- If some $\lambda_i = 0$, then 0 is an eigenvalue of A^k ;
- for $k = 0$ we get $A^0 = I$ with eigenvalues all equal to 1 and the same eigenvectors v_i .

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Reminder: Endomorphisms

Definition

An endomorphism f of a vector space E is a linear map from the space to itself:

$$f : E \rightarrow E.$$

NB: we denote by $\mathcal{L}(E)$ the set of endomorphisms on E .

Finite-dimensional case

- When E is finite-dimensional, the study of f reduces to the study of its matrix with respect to a chosen basis.
- The resulting matrix is square; often the same basis of E is used at the source and at the target.

Reduction of endomorphisms

In linear algebra, the **reduction** of an endomorphism aims to express matrices and endomorphisms in a simpler form, for example to make computations easier.

Method

Reduction essentially consists in decomposing the vector space as a direct sum of invariant subspaces on which the induced endomorphism is simpler:

$$E = U_1 \oplus \cdots \oplus U_k \quad \text{with } f(U_i) \subseteq U_i.$$

NB: Geometrically, this amounts to choosing a basis of the space in which the endomorphism has a simple expression (e.g., block or diagonal form).

Diagonalization

- In finite dimension, diagonalizing an endomorphism means finding a basis in which the matrix of f is diagonal.
- Not every endomorphism is diagonalizable; in some cases one can at best triangularize it, i.e. put it in upper-triangular form.
- Diagonalization is useful for analyzing f , computing powers f^k , searching for square roots of f , etc.

But why do we care ??

Reminder: eigenvectors as stable directions

Let $A \in \mathbb{R}^{d \times d}$ be a linear transformation acting on feature vectors:

$$x \mapsto Ax.$$

An eigenvector $v \neq 0$ satisfies

$$Av = \lambda v.$$

Interpretation: the direction of v is preserved by the transformation. Only its magnitude is changed by the factor λ .

- If $|\lambda| > 1$, the direction is amplified.
- If $|\lambda| < 1$, the direction is contracted.
- If $\lambda < 0$, the orientation is reversed.

In ML, this viewpoint is useful for analyzing repeated transformations, and recurrent neural networks for example.

ML illustration: repeated transformations and stability

Consider an iterative linear system

$$x_{t+1} = Ax_t.$$

If A is diagonalizable, $A = PDP^{-1}$, then

$$A^t = PD^tP^{-1}.$$

Therefore the long-term behavior is controlled by the eigenvalues:

$$D^t = \text{diag}(\lambda_1^t, \dots, \lambda_d^t).$$

- If $|\lambda_i| < 1$, the corresponding direction disappears.
- If $|\lambda_i| > 1$, the corresponding direction explodes.
- If $|\lambda_i| = 1$, the corresponding direction is preserved in magnitude.

NB: This is a simplified intuition for stability questions in recurrent models and iterative algorithms.

ML illustration: spectral radius and exploding gradients

The **spectral radius** of a matrix A is

$$\rho(A) = \max_{\lambda \in \text{Sp}(A)} |\lambda|.$$

For repeated applications of a matrix,

$$x_t = A^t x_0,$$

the spectral radius gives information about stability.

- If $\rho(A) < 1$, powers of A tend to contract vectors.
- If $\rho(A) > 1$, some directions may grow exponentially.

ML interpretation: in recurrent neural networks, repeated multiplication by weight matrices may cause vanishing or exploding signals and gradients.¹

¹Further reading: R. Pascanu, T. Mikolov, Y. Bengio, *On the difficulty of training Recurrent Neural Networks*, arXiv:1211.5063. <https://arxiv.org/abs/1211.5063> 

ML illustration: PageRank as an eigenvector problem

The PageRank algorithm represents the web as a directed graph.

Let P be a transition matrix, where P_{ij} describes the probability of moving from page j to page i .

A stationary importance vector r satisfies

$$Pr = r.$$

Thus r is an eigenvector of P associated with the eigenvalue 1.

Interpretation: the most important pages are found by solving an eigenvector problem.

More generally, eigenvectors can describe stable states of Markov chains and random walks on graphs.²

²read more at <https://en.wikipedia.org/wiki/PageRank>        

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Eigenvalue, eigenvector

Let E be a vector space defined over a field $\mathbb{K}$. For an endomorphism $f \in \mathcal{L}(E)$, there *may* be vectors which, when f is applied to them, are simply scaled by some constant.

Definition

We say that a nonzero vector $x \in E$ is an **eigenvector** of f corresponding to the **eigenvalue** λ if

$$f(x) = \lambda x$$

NB: The zero vector is excluded from this definition because $f(\mathbf{0}) = \mathbf{0} = \lambda \mathbf{0}$ for every $\lambda \in \mathbb{K}$.

Eigenspace and eigenspectrum

Definition (Eigenspectrum)

The **eigenspectrum** (or **spectrum**) of an endomorphism is the set of its eigenvalues. We denote it $\text{Sp}(f)$.

$$\text{Sp}(f) = \{\lambda \in \mathbb{K}, \exists v \in E, v \neq 0_E \text{ and } f(v) = \lambda v\}$$

NB: When E is finite-dimensional, eigenvalues also have algebraic multiplicities, defined later using the characteristic polynomial. We will denote m_λ the multiplicity of eigenvalue λ , writing the eigenspectrum as a set and specifying the multiplicity of each eigenvalue separately.

Eigenspace

Definition (Eigenspace)

The set of all eigenvectors of f corresponding to the same eigenvalue λ , together with the zero vector, is called an **eigenspace**.

Notation: Let $\lambda \in \mathbb{K}$ be an eigenvalue of f . We denote the eigenspace of f associated to the eigenvalue λ by $E_\lambda(f)$ or $\text{Eig}(f, \lambda)$.

$$E_\lambda(f) = \{v \in E, f(v) = \lambda v\}$$

Can you express $E_\lambda(f)$ as the kernel of a linear application ?

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$$E_\lambda(f) = \text{Ker}(f - \lambda \text{Id})$$

What can you say about the dimension of this subspace ?

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$$\dim(E_\lambda(f)) \geq 1$$

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$$E_\lambda(f) = \text{Ker}(f - \lambda \text{Id})$$

What can you say about the dimension of this subspace ?

$$\dim(E_\lambda(f)) \geq 1$$

We will see that

$$\dim(E_\lambda(f)) \leq m_\lambda$$

Manipulating eigenvalues

Proposition

Let $\mathbf{x}$ be an eigenvector of $A \in \mathcal{M}_n(\mathbb{K})$ with eigenvalue λ . Then:

- i For any $\gamma \in \mathbb{K}$, $\mathbf{x}$ is an eigenvector of $A + \gamma I_n$ with eigenvalue $\lambda + \gamma$.
- ii If A is invertible, then $\lambda \neq 0$ and $\mathbf{x}$ is an eigenvector of A^{-1} with eigenvalue λ^{-1} .
- iii If A is invertible, then $A^k \mathbf{x} = \lambda^k \mathbf{x}$ for any $k \in \mathbb{Z}$.

Manipulating eigenvalues

Proof

(i) By computing

$$(A + \gamma \mathbf{I})x = Ax + \gamma \mathbf{I}x = \lambda x + \gamma x = (\lambda + \gamma)x$$

(ii) Suppose A is invertible. Then

$$x = A^{-1}Ax = A^{-1}(\lambda x) = \lambda A^{-1}x$$

Dividing by λ , which is valid because the invertibility of A implies $\lambda \neq 0$, gives $\lambda^{-1}x = A^{-1}x$.

(iii) Cf previous exercise. The case $k \geq 0$ follows by induction on k . Then the general case $k \in \mathbb{Z}$ follows by combining the $k \geq 0$ case with (ii).

Properties

Proposition

Let $f \in \mathcal{L}(E)$ an endomorphism. Then 0 is an eigenvalue of f if and only if $\ker(f) \neq \{0_E\}$ i.e. f is not injective.

$$0 \in \text{Sp}(f) \iff \ker(f) \neq \{0_E\} \iff f \text{ is not injective.}$$

NB: directly follows from expressing $E_\lambda(f)$ as $\ker(f - \lambda \text{Id})$.

Determinant and eigenspace

Let $f \in \mathcal{L}(E)$ an endomorphism, with E a finite-dimensional vector space. Then

$$\lambda \in \text{Sp}(f) \iff \det(\lambda \text{Id}_E - f) = 0.$$

Proof: exercise

Properties

Let's prove the property on the determinant of the linear map $\lambda Id_E - f$.
($\implies$) Let $\lambda \in \text{Sp}(f)$. Assume there exists $u \in E$ with $u \neq 0$ such that $f(u) = \lambda u$. Then

$$\lambda u - f(u) = 0 \iff (\lambda Id_E - f)(u) = 0 \iff u \in \ker(\lambda Id_E - f).$$

Hence $\lambda Id_E - f$ is not bijective (i.e. its kernel contains a nonzero vector), and therefore

$$\det(\lambda Id_E - f) = 0.$$

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Hence $\lambda Id_E - f$ is not bijective (i.e. its kernel contains a nonzero vector), and therefore

$$\det(\lambda Id_E - f) = 0.$$

($\impliedby$) If $\det(\lambda Id_E - f) = 0$, then $\lambda Id_E - f$ is not bijective, hence not injective (since E is finite-dimensional).

Thus there exists $u \in \ker(\lambda Id_E - f)$ with $u \neq 0$, which means $f(u) = \lambda u$. Therefore λ is an eigenvalue of f (i.e., $\lambda \in \text{Sp}(f)$).

Proposition

Let $f \in \mathcal{L}(E)$ and let $\lambda_1, \dots, \lambda_p$ be p *pairwise-distinct* eigenvalues of f .

- 1 The sum of subspaces $E_{\lambda_1}(f) + \dots + E_{\lambda_p}(f)$ is a direct sum.
- 2 If $(u_1, u_2, \dots, u_p) \in E^p$ such that

$$\forall i \in \llbracket 1, p \rrbracket, u_i \neq 0, \text{ and } f(u_i) = \lambda_i u_i$$

Then the family $(u_1, u_2, \dots, u_p)$ is linearly independent.

Proof of (1): by induction. Proof of (2): follows from (1) with the definition of a direct sum.

Proof of the proposition – Direct sum

(1) We prove by induction on p that the following sum of subspaces is a direct sum.

$$E_{\lambda_1}(f) + \cdots + E_{\lambda_p}(f)$$

For $p = 1$, the result is immediate. Then for some $p > 1$, assume it holds for $p - 1$, and let

$$u_1 + \cdots + u_p = 0, \quad u_i \in E_{\lambda_i}(f).$$

Applying the map $f - \lambda_p \text{Id}$ to this eq. gives

$$\sum_{i=1}^{p-1} (\lambda_i - \lambda_p) u_i = 0.$$

Since $(\lambda_i - \lambda_p) u_i \in E_{\lambda_i}(f)$, the induction hypothesis yields

$$(\lambda_i - \lambda_p) u_i = 0 \quad (i = 1, \dots, p-1).$$

The eigenvalues being pairwise distinct, $\lambda_i - \lambda_p \neq 0$, hence $u_1 = \cdots = u_{p-1} = 0$, and therefore also $u_p = 0$.

Thus $E_{\lambda_1}(f) + \cdots + E_{\lambda_p}(f) = E_{\lambda_1}(f) \oplus \cdots \oplus E_{\lambda_p}(f)$.

Proof of the proposition – Linear independence

(2) Let

$$a_1 u_1 + \cdots + a_p u_p = 0.$$

Since

$$a_i u_i \in E_{\lambda_i}(f)$$

and the sum of the eigenspaces is direct by (1), we must have

$$a_i u_i = 0 \quad \text{for every } i.$$

Since $u_i \neq 0$, it follows that $a_i = 0$ for every i .

Hence

$$(u_1, \dots, u_p)$$

is linearly independent.

Properties

Cardinal of the eigenspectrum

If E is a finite-dimensional vector space, with $\dim(E) = n$, and $f \in \mathcal{L}(E)$, then

$$\text{Card}(\text{Sp}(f)) \leq n$$

Proof.

Properties

Cardinal of the eigenspectrum

If E is a finite-dimensional vector space, with $\dim(E) = n$, and $f \in \mathcal{L}(E)$, then

$$\text{Card}(\text{Sp}(f)) \leq n$$

Proof. Suppose $\text{Card}(\text{Sp}(f)) \geq n + 1$, with distinct eigenvalues. Using the previous property, we would have a linearly independent family of vectors $(u_1, u_2, \dots, u_n, u_{n+1})$ associated with eigenvalues $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$. But $\dim(E) = n$. Absurd.

Properties (bonus exercise)

Proposition

Let $f, g \in \mathcal{L}(E)$ two endomorphisms that commute. Then the eigenspaces of f are stable by g .

NB: Let $\lambda \in \text{Sp}(f)$. This proposition writes :

$$g(E_\lambda(f)) \subseteq E_\lambda(f)$$

Since f and g are commutative, $\lambda \text{Id}_E - f$ and g are commutative too. Writing $E_\lambda(f)$ as $\ker(\lambda \text{Id}_E - f)$, the eigenspace of f associated to λ is stable by g .

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Characteristic polynomial

We consider a finite-dimensional vector space E , $\dim(E) = n$.

Definition

Let $f \in \mathcal{L}(E)$. We call characteristic polynomial of f the polynomial, denoted χ_f , defined as

$$\chi_f(X) = \det(X\text{Id}_E - f) \in \mathbb{K}[X]$$

NB: alternative notation $P_\lambda(f)$; also written $\chi_f(\lambda)$ instead of $\chi_f(X)$, with λ a variable.

Similarly, for a square matrix $A \in \mathcal{M}_n(\mathbb{K})$, we will denote $\chi_A(X) = \det(XI_n - A)$.

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What is the degree of χ_f ? What is the value of its leading coefficient ?

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What is the degree of χ_f ? What is the value of its leading coefficient ?
The leading coefficient is 1, we say the polynomial is **monic**, of degree n .

Characteristic polynomial

- Following the property on the determinant of the linear map $\lambda \text{Id}_E - f$, a value λ is an eigenvalue of f if and only if it is a root of its characteristic polynomial.

$$\lambda \in \text{Sp}(f) \iff \chi_f(\lambda) = 0, \quad \chi_f(\lambda) = \det(\lambda \text{Id}_E - [f]_B).$$

- $\chi_A(\lambda)$ is a polynomial of degree n .
- Remark that the leading coefficient of $\chi_A(\lambda)$ is 1 (unitary/monic polynomial).
- If A is the matrix of f in a given basis of E , then we have the equivalence

$$\chi_A = \chi_f$$

This means that the characteristic polynomial of a linear map does not depend on the choice of basis. Like the determinant, it does not vary under matrix similarity (NB: FR “invariant de similitude”)

Multiplicity of an eigenvalue

Definition

Let $f \in \mathcal{L}(E)$, $A \in \mathcal{M}_n(\mathbb{K})$, and $\lambda \in \text{Sp}(f)$. We call **multiplicity** of an eigenvalue λ its multiplicity as a root of the characteristic polynomial of f (or of A , respectively). It is denoted m_λ .

$$m_\lambda = \max \left\{ k \in \mathbb{N}^* \mid (X - \lambda)^k \text{ divides } \chi_f(X) \right\}.$$

NB: For any $\lambda \in \text{Sp}(f)$,

$$1 \leq m_\lambda \leq n$$

Proposition

Let $\lambda \in \text{Sp}(f)$. We have $\dim(E_\lambda(f)) \leq m_\lambda$. Similarly, for a square matrix $A \in \mathcal{M}_n(\mathbb{K})$, $\dim(E_\lambda(A)) \leq m_\lambda$.

Multiplicity of an eigenvalue

Not always equal

The dimension of the eigenspace associated to a given eigenvalue λ is not necessarily equal to the multiplicity of this eigenvalue in the characteristic polynomial.

Example

Example : take the matrix A with zeros everywhere, except on its diagonal and upper diagonal, filled with ones. $\text{Sp}(A) = \{1\}$, $\chi_A(\lambda) = (\lambda - 1)^n$. $E_1(A) = \text{span}((1, 0, \dots, 0)^\top)$. is of dimension 1, whereas the multiplicity of 1 as root of the characteristic polynomial of A is n .

NB: Equality holds if $m_\lambda = 1$.

Not all matrices are diagonalizable

Example

$$A = \begin{bmatrix} 1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 1 & \dots & 0 & 0 \\ \dots & & & \dots & & \\ 0 & 0 & 0 & \dots & 1 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

Not all matrices are diagonalizable

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$\text{Sp}(A) = \{1\}$. If A was diagonalizable, then A would be similar to the identity matrix. i.e. $D = \text{diag}(1, \dots, 1) = I_n$.

But $PI_nP^{-1} = I_n$, so I_n is the only matrix similar to I_n .

This leaves us with $A = I_n$, absurd. So A is not diagonalizable.

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This leaves us with $A = I_n$, absurd. So A is not diagonalizable.

We can generalize this example: if a square matrix M is diagonalizable and has a unique eigenvalue λ , then M is similar to λI_n . But

$$P(\lambda I_n)P^{-1} = \lambda I_n,$$

so in fact $M = \lambda I_n$. In particular, if $\lambda = 1$, then $M = I_n$.

Recap - Characteristic polynomial

- The **characteristic polynomial** of an endomorphism of a finite-dimensional vector space is the characteristic polynomial of the matrix of that endomorphism over any basis.
- This means that the characteristic polynomial does not depend on the choice of a basis.
- The characteristic polynomial of a square matrix is a polynomial which is invariant under matrix similarity
- It has the eigenvalues of this matrix as roots.
- The **characteristic equation** is the equation obtained by equating the characteristic polynomial to zero.

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Diagonalizable matrix

In this section we consider a finite-dimensional vector space E , $\dim(E) = n$.

Definition

- We say that $f \in \mathcal{L}(E)$ is **diagonalizable** if there exists a basis $\mathcal{B}$ of E formed by eigenvectors of f , i.e. $[f]_{\mathcal{B}}$ the matrix of f in $\mathcal{B}$ is diagonal.
- We say that $A \in \mathcal{M}_n(\mathbb{K})$ is **diagonalizable** if A is similar to a diagonal matrix, i.e. there exists an invertible matrix $P \in \mathcal{M}_n(\mathbb{K})$ and a diagonal matrix D such that

$$A = PDP^{-1}$$

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$$A = PDP^{-1}$$

NB : “Diagonalizing” an endomorphism means finding a basis of E in which the matrix representation of f is diagonal.

“Diagonalizing” a matrix $A \in \mathcal{M}_n(\mathbb{K})$ means finding an invertible matrix $P \in \mathcal{M}_n(\mathbb{K})$ and a diagonal matrix $D \in \mathcal{M}_n(\mathbb{K})$ such that $A = PDP^{-1}$.

Properties

Let $f \in \mathcal{L}(E)$.

Proposition

- Let $\mathcal{B}$ be a basis such that $[f]_{\mathcal{B}} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then $\text{Sp}(f) = \{\lambda_1, \dots, \lambda_n\}$.
- The following statements are equivalent.
 - ① f is diagonalizable
 - ② There exists a basis $\mathcal{B}$ of E such that $[f]_{\mathcal{B}}$ is diagonal.
 - ③ For any basis $\mathcal{B}$ of E , $[f]_{\mathcal{B}}$ is diagonalizable.

NB : (1) $\chi_f(\lambda) = \chi_D(\lambda)$ with $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Since $\chi_D(\lambda) = \prod_{i=1}^n (\lambda - \lambda_i)$ we get $\text{Sp}(f) = \{\lambda_1, \dots, \lambda_n\}$.
(2) use property on similar matrices.

Equivalent statements on diagonalizability

Proposition

Let $f \in \mathcal{L}(E)$. Then the following statements are equivalent.

- ① f is diagonalizable
- ② $\dim(E) = \sum_{\lambda \in \text{Sp}(f)} \dim(E_\lambda(f))$
- ③ $E = \bigoplus_{\lambda \in \text{Sp}(f)} E_\lambda(f)$.

Equivalent statements on diagonalizability

Recall that the eigenspaces are in direct sum. Hence

$$\dim\left(\bigoplus_{\lambda \in \text{Sp}(f)} E_{\lambda}(f)\right) = \sum_{\lambda \in \text{Sp}(f)} \dim(E_{\lambda}(f))$$

and the equivalence between (2) and (3).

If we suppose that (1) f is diagonalizable, then there exists a basis $\mathcal{B}$ of E formed by eigenvectors of f . So $\bigoplus_{\lambda \in \text{Sp}(f)} E_{\lambda}(f)$ contains a basis of E . It follows that $\sum_{\lambda \in \text{Sp}(f)} E_{\lambda}(f) = E$ (3).

Now suppose (3): we have $E = \bigoplus_{\lambda \in \text{Sp}(f)} E_{\lambda}(f)$. So there exists a basis of E adapted to this decomposition of E . This basis is formed by eigenvectors of f ... So f is diagonalizable (1).

Split Polynomial

Definition

Let $\mathbb{K}$ be a field. A non-constant polynomial $P \in \mathbb{K}[X]$ *splits over* $\mathbb{K}$ if

$$P(X) = C \prod_{j=1}^n (X - a_j),$$

where each $a_j \in \mathbb{K}$ and $C \in \mathbb{K}^*$. Equivalently, P can be written as a product of degree-1 polynomials with coefficients in $\mathbb{K}$. In this case, C is called the **leading coefficient** of P .

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Dependence on the Field

The splitting property depends on $\mathbb{K}$. For example,

$$X^2 + 1 = (X - i)(X + i) \quad \text{splits over } \mathbb{C},$$

but it does not split over $\mathbb{R}$.

Simple Roots

Definition

A split polynomial P has *simple roots* if, in the previous factorization, the numbers $a_1, \dots, a_n$ are pairwise distinct (i.e., each root has multiplicity 1).

Example

Over $\mathbb{K} = \mathbb{Q}$ Consider

$$P(X) = 5(X - 1)(X + 2)(X - 3) \in \mathbb{Q}[X].$$

All roots $1, -2, 3$ lie in $\mathbb{Q}$, so P splits over $\mathbb{K}$ with leading coefficient 5. The roots are pairwise distinct, hence P is split with simple roots.

NB: Expanding,

$$P(X) = 5(X^3 - 2X^2 - 5X + 6) = 5X^3 - 10X^2 - 25X + 30.$$

Characteristic polynomial and Diagonalizability

Theorem (for endomorphisms)

Let E be a vector space defined over a field $\mathbb{K}$, $\dim(E) = n$ and $f \in \mathcal{L}(E)$. The endomorphism f is diagonalizable if and only if

- 1 Its characteristic polynomial χ_f is split.
- 2 For each $\lambda \in \text{Sp}(f)$, $\dim(E_\lambda(f)) = m_\lambda$.

Theorem (for matrices)

Let $A \in \mathcal{M}_n(\mathbb{K})$. It is diagonalizable if and only if

- 1 Its characteristic polynomial χ_A is split.
- 2 For each $\lambda \in \text{Sp}(A)$, $\dim(E_\lambda(A)) = m_\lambda$, or equivalently,

$$\text{rank}(\lambda I_n - A) = n - m_\lambda$$

Annihilating Polynomials

Definition

Let E be a finite-dimensional $\mathbb{K}$ -vector space with $\dim E = n$. Let $f \in \mathcal{L}(E)$ be an endomorphism and let $P \in \mathbb{K}[X]$ be a polynomial. P is called an *annihilating polynomial* for f if

$$P(f) = 0_{\mathcal{L}(E)} \quad (\text{i.e., the zero endomorphism}).$$

Notation reminder: If $P(X) = a_0 + a_1X + \cdots + a_mX^m$, then $P(A) = a_0I + a_1A + \cdots + a_mA^m$.

Eigenvalues and Annihilating Polynomials

Theorem (Spectral Mapping for Polynomials)

Let $f : E \rightarrow E$ be linear map over a field $\mathbb{K}$ and let $P \in \mathbb{K}[X]$ satisfy $P(f) = 0$, i.e. P is an annihilating polynomial for f . Then every eigenvalue λ of f is a root of P :

$$\text{Sp}(f) \subseteq \{z \in \mathbb{K} : P(z) = 0\}.$$

Warning

The eigenspectrum of f is included in the set of roots of an annihilating polynomial P .

Eigenvalues and Annihilating Polynomials

Proof. Let $\lambda \in \mathbb{K}$ be an eigenvalue of f and choose $v \in E$, $v \neq 0_E$ with $f(v) = \lambda v$.

We write the polynomial as $P(X) = \sum_{k=0}^m a_k X^k$ (with $a_k \in \mathbb{K}$). Note that $f^0 = \text{Id}_E$.

By induction, $f^k(v) = \lambda^k v$ for all $k \geq 0$. Then

$$P(f)(v) = \sum_{k=0}^m a_k f^k(v) = \sum_{k=0}^m a_k \lambda^k v = P(\lambda) v.$$

Since $P(f) = 0$ by hypothesis, we have $0 = P(f)(v) = P(\lambda) v$.

Because $v \neq 0$, it follows that $P(\lambda) = 0$. Hence every eigenvalue λ of f is a root of P .

Minimal Polynomial of a Matrix

Definition

Let $A \in M_n(\mathbb{K})$. The **minimal polynomial** of A , denoted μ_A , is the unique *monic* polynomial of least degree such that

$$\mu_A(A) = 0.$$

Corollary (corollary of the previous theorem)

*In particular, all eigenvalues of A are roots of μ_A , the minimal polynomial of A .
NB: Equivalently, μ_A divides every annihilating polynomial of A (in particular, following Cayley-Hamilton, $\mu_A \mid \chi_A$).*

Example

$$B = \text{diag}(1, 2, 2).$$

Then $(B - I)(B - 2I) = 0$, while $B - I \neq 0$ and $B - 2I \neq 0$. Thus

$$\mu_B(X) = (X - 1)(X - 2).$$

Cayley-Hamilton Theorem

This theorem tells us that the characteristic polynomial of a given endomorphism is an annihilating polynomial for this endomorphism.

Theorem (Cayley-Hamilton)

For any $n \times n$ matrix A over a field K , let

$$\chi_A(\lambda) = \det(\lambda I_n - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$$

be its characteristic polynomial. Then

$$\chi_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1A + c_0I_n = 0.$$

NB: $\chi_A(A) = 0$ means that the matrix A satisfies its own characteristic polynomial.

Cayley-Hamilton Theorem

Note on the **Existence of an Annihilating Polynomial**

Since $\dim \mathcal{L}(E) = n^2$, the family

$$(\text{Id}_E, f, f^2, \dots, f^{n^2})$$

of $n^2 + 1$ endomorphisms is linearly dependent. Hence there exist $a_0, \dots, a_{n^2} \in \mathbb{K}$, not all zero, such that

$$a_0 \text{Id}_E + a_1 f + \dots + a_{n^2} f^{n^2} = 0.$$

Setting $P(X) = a_0 + a_1 X + \dots + a_{n^2} X^{n^2}$ yields a nonzero polynomial with $P(f) = 0$.

Example: Characteristic polynomial in 2×2

Computation (using $\chi_A(\lambda) = \det(\lambda I_2 - A)$) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$\lambda I_2 - A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} - \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix}.$$

Hence

$$\begin{aligned} \chi_A(\lambda) &= \det \begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix} = (\lambda - a)(\lambda - d) - (-b)(-c) \\ &= \lambda^2 - (a + d)\lambda + (ad - bc). \end{aligned}$$

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Identification

Since $\text{tr}(A) = a + d$ and $\det(A) = ad - bc$, we obtain

$$\chi_A(\lambda) = \lambda^2 - (\text{tr } A) \lambda + \det(A).$$

Example: Characteristic polynomial in 2×2

Now let us suppose A of size 2×2 is invertible. Using the previous formula, find an explicit formula for the inverse of A .

Cayley-Hamilton theorem - Examples

Example

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad \chi_A(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6.$$

Compute $A^2 = \begin{pmatrix} 4 & 5 \\ 0 & 9 \end{pmatrix}$. Then

$$A^2 - 5A + 6I_2 = \begin{pmatrix} 4 & 5 \\ 0 & 9 \end{pmatrix} - \begin{pmatrix} 10 & 5 \\ 0 & 15 \end{pmatrix} + \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} = 0.$$

Bonus: Compute the inverse from Cayley-Hamilton. $\det(A) = 6$, $\text{tr}(A) = 5$, hence

$$A^{-1} = \frac{1}{6}(5I_2 - A) = \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{3} \end{pmatrix}.$$

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Triangularizability

Definition

- Let $f \in \mathcal{L}(E)$. We say that f is **triangularizable** if there exists a basis $\mathcal{B}$ of E such that $[f]_{\mathcal{B}}$ the matrix of f in $\mathcal{B}$, is triangular.
- Let $A \in \mathcal{M}_n(\mathbb{K})$. We say that A is **triangularizable** if A is similar to a triangular matrix.

NB: we often consider upper triangular matrices.

If $f \in \mathcal{L}(E)$, and $\mathcal{B} = (e_1, e_2, \dots, e_n)$ is a basis of E , then $\mathcal{M}_{\mathcal{B}}(f)$ is upper triangular if and only if

$$\forall i \in \llbracket 1, n \rrbracket, \quad f(e_i) \in \text{span}(e_1, \dots, e_i).$$

Warning

A diagonalizable matrix is triangularizable, but not reciprocally.

Triangularizability

Triangularizing a linear map f means finding a basis $\mathcal{B}$ such that $[f]_{\mathcal{B}}$ is triangular.

Triangularizing a matrix $A \in \mathcal{M}_n(\mathbb{K})$ consists in finding P invertible and T triangular such that $A = PTP^{-1}$.

Theorem (Criterion of Triangularisability)

Let $f \in \mathcal{L}(E)$. Then f is triangularizable if and only if its characteristic polynomial χ_f is split. (FR: scindé) Similarly, a square matrix A is triangularizable if and only if its characteristic polynomial χ_A is split.

Proof : by induction. NB : all matrices of $\mathcal{M}_n(\mathbb{C})$ are triangularizable because all the polynomials from $\mathbb{C}[X]$ are split, but all real valued matrices are not triangularizable.

Triangularizability, trace and determinant

Proposition

Assume that χ_f splits over $\mathbb{K}$:

$$\chi_f(X) = \prod_{\lambda \in \text{Sp}(f)} (X - \lambda)^{m_\lambda}.$$

Then:

- ① $\sum_{\lambda \in \text{Sp}(f)} m_\lambda = n.$
- ② $\det(f) = \prod_{\lambda \in \text{Sp}(f)} \lambda^{m_\lambda}.$
- ③ $\text{tr}(f) = \sum_{\lambda \in \text{Sp}(f)} m_\lambda \lambda.$

Method - Triangularisability

Let $A \in \mathcal{M}_n(\mathbb{K})$ such that χ_A is split over $\mathbb{K}$,

$$\chi_A(X) = \prod_{i=1}^p (X - \lambda_i)^{m_i}$$

where $\lambda_1, \dots, \lambda_p$ are the distinct eigenvalues, with multiplicity $m_1, m_2, \dots, m_p$. NB : we know that $\sum_{i=1}^p m_i = n$.

For each $i \in \llbracket 1, p \rrbracket$ we look for a basis $\mathcal{B}_i$ of $\ker(\lambda_i I_n - A)$.

If $\text{Card}(\mathcal{B}_i) = m_i$ (i.e. $\dim(\ker(\lambda_i I_n - A)) = m_i$) then we can diagonalize.

Otherwise, we look for vectors to add to $\mathcal{B}_i$ to make it a basis of $\ker((\lambda_i I_n - A)^2)$, etc. i.e. generalized eigenvectors, i.e. vectors in

$$\ker((A - \lambda_i I_n)^r)$$

for increasing values of r .

Finally, the family of vectors $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \dots \mathcal{B}_p$ is a basis where A can be triangularized.

Nilpotent matrices

Definition

A **nilpotent** matrix N is a square matrix such that there exists a positive integer k such that

$$N^k = 0$$

NB: the smallest power k for which N^k is the zero matrix is sometimes called the **index** of N .

Example

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

N is nilpotent with index 2, since $N^2 = 0$.

We can prove that any upper triangular matrix $T \in \mathcal{M}_n(\mathbb{K})$ with zeros on the diagonal is nilpotent, with index at most n .

Nilpotent matrices

Example

$$B = \begin{bmatrix} 0 & 2 & 1 & 6 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B^3 = \begin{bmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B^4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore B is nilpotent, with index 4.

Nilpotent matrices

What to use nilpotent matrices for ?

Example

Triangularisability then decomposition of an upper triangular matrix as the sum of a diagonal matrix and of a nilpotent matrix.

NB: The determinant and trace of a nilpotent matrix are always zero. So a nilpotent matrix cannot be invertible.

Cayley-Hamilton theorem - Examples

Example (Jordan block)

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \chi_J(\lambda) = (\lambda - 1)^3.$$

$$\chi_J(J) = 0 \iff (J - I_3)^3 = 0 \iff J^3 - 3J^2 + 3J - I_3 = 0.$$

Writing $N = J - I_3$ (nilpotent with $N^3 = 0$), any power J^m reduces to a polynomial of degree ≤ 2 in N :

$$J^m = (I_3 + N)^m = I_3 + mN + \binom{m}{2}N^2 \quad (N^3 = 0).$$

This illustrates how we can use C-H to compute powers of a matrix even when the matrix is not diagonalizable.

Practical implications of Cayley-Hamilton theorem

- **Reduce powers.** Any A^m ($m \geq n$) can be rewritten as a linear combination of $I, A, \dots, A^{n-1}$ using $\chi_A(A) = 0$.
- **Compute inverses.** If $\det(A) \neq 0$, C-H yields a polynomial expression for A^{-1} in terms of $I, A, \dots, A^{n-1}$ (e.g., explicit in the 2×2 example we saw previously).
- **Minimal polynomial.** By Cayley-Hamilton, χ_A is an annihilating polynomial, so the minimal polynomial μ_A divides χ_A .
- **Linear recurrences.** For vectors $x_k = A^k x_0$, the sequence satisfies the recurrence with coefficients from χ_A (useful in control theory... or in deep learning!).

Example exercise

Example

Consider the following matrix, with $a > 0$:

$$A = \begin{bmatrix} -1 & a & -a \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

- 1 Compute χ_A and show that A is not diagonalizable.
- 2 Find v_1, v_2, v_3 in $\mathcal{M}_{3,1}(\mathbb{R})$ such that $Av_1 = -v_1$, $Av_2 = v_1 - v_2$, $Av_3 = v_1 + v_2 - v_3$.

- 3 Show that A is similar to $T = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

- 4 Use a nilpotent matrix to compute A^n for any $n \in \mathbb{N}$.

Next class

Review the [method](#) !

Reminder : you can find the eigenspace $E_\lambda(A)$ of $A \in \mathcal{M}_n(\mathbb{K})$ associated with an eigenvalue λ by solving the linear system resulting from $AX = \lambda X$. If you need practice for finding the *set of solutions* of a linear system please refer to Example 2.2 from the MML book.

Next class : Inner product spaces, normed spaces, spectral theorem.
You can prepare by reading the MML book, Chapter 4.